

CoM-HOI: Belief-Guided Cooperative Humanoid-Object Interactions under Uncertain Mass Properties

Anonymous submission

Abstract

Existing cooperative humanoid transport methods largely coordinate agents based on visible object geometry. However, safe carrying fundamentally requires aligning the support configuration with the object’s center of mass (CoM), which may deviate substantially from its geometric center and cannot be reliably inferred from appearance alone. We present CoM-HOI, a belief-guided hierarchical framework for cooperative transport under uncertain mass properties. CoM-HOI treats the first lift as interaction-based physical probing: humanoids perform a low-amplitude lift from a geometry-based formation, while a physics-based estimator converts short-horizon interaction responses into a probabilistic belief over the hidden CoM. Conditioned on this belief, a hierarchical manager predicts load distribution, evaluates support stability and per-agent load feasibility, and triggers safe putdown and team re-formation when necessary. The framework further combines belief-biased contact allocation, CoM-aware formation initialization, and analytic load-aware rewards. Experiments on hidden-eccentric transport tasks show that CoM-HOI remains competitive with geometry-based coordination under mild eccentricity and, under larger CoM offsets, maintains robust transport while achieving increasingly pronounced gains over existing methods.

1 Introduction

Humanoid robots are expected to assist humans in everyday environments, where many tasks require coordinated whole-body interaction with large or heavy objects (Tuci, Alkilabi, and Akanyeti 2018; Merel et al. 2020; Han et al. 2024; Sui et al. 2026). Cooperative transport is a representative example: multiple humanoids must jointly approach, contact, lift, stabilize, and move an object beyond the capability of a single agent. Recent advances in physics-based humanoid control and cooperative manipulation have increasingly integrated locomotion, manipulation, contact regulation, and multi-agent coordination. Model-based and learning-based approaches now support contact-rich whole-body skills, coordinated force interaction, and team-based transport of objects beyond individual capabilities (Suomalainen, Karayianidis, and Kyrki 2022; Dharmendrasinh et al. 2024; Gu et al.

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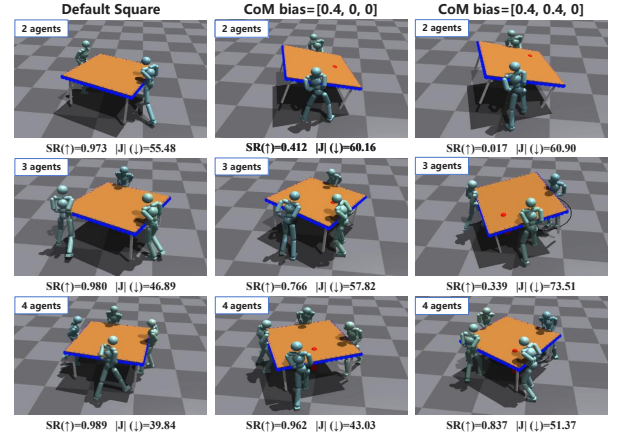


Figure 1: Performance under different CoM biases and team sizes. Geometry-based coordination performs well for symmetric objects but degrades under hidden CoM shifts, particularly with fewer agents.

2026). These developments establish a foundation for physically grounded multi-humanoid cooperation.

However, existing cooperative humanoid-object interaction (HOI) formulations largely assume visible geometry suffices for stable support formation (Vianello et al. 2021; Bai et al. 2025). Agents are commonly distributed around the geometric center or object boundary. This assumption becomes unreliable for objects with unknown and eccentric mass properties. A visually symmetric object may contain a hidden heavy component, shifting its CoM away from the geometric center (Robinson et al. 2023; Cao 2025). Consequently, a geometrically balanced formation may lift the object without visible tipping while imposing excessive vertical load on agents near the hidden CoM. This exposes an underexplored distinction in cooperative transport: safe interaction requires not only preventing object tilt and contact loss, but also maintaining per-agent load feasibility.

This paper studies cooperative humanoid-object transport under uncertain mass properties, where a humanoid team must infer the latent mass distribution of an object with an unknown and potentially eccentric CoM through interaction and adapt its support formation for safe transport.

We propose **CoM-HOI**, a belief-guided hierarchical framework for cooperative humanoid-object interactions under uncertain mass properties. CoM-HOI treats the first lift as interaction-based physical probing: from a geometry-based support configuration, the team performs a low-amplitude lift and brief hold that expose force, contact, and object-response cues to the hidden mass distribution. A physics-based estimator aggregates these cues into a probabilistic CoM belief and predicts per-agent loads under quasi-static balance. The hierarchical manager uses this belief to assess support stability and load feasibility, retaining the formation when safe and commanding a controlled putdown before re-formation otherwise. Re-formation biases candidate contacts toward the estimated heavy side while preserving boundary coverage and shifts the re-lift formation toward the estimated CoM. Analytic load-aware and stability rewards guide agent-wise load sharing and suppress excessive tilt, rotation, and slip. Together, these components preserve inherited whole-body carrying skills while enabling decentralized adaptation to hidden mass properties.

We evaluate CoM-HOI on hidden-eccentric transport tasks with unchanged visible geometry across three object shapes and eccentricity levels, with randomized CoM directions. CoM-HOI remains competitive with geometry-based coordination under mild eccentricity and achieves increasingly pronounced advantages at larger offsets while maintaining robust transport. Estimation and ablation analyses further examine belief reliability and downstream adaptation, supporting interaction-based physical probing and belief-guided re-formation as complements to geometric coordination.

Our contributions are summarized as follows:

- We formulate cooperative humanoid-object transport under unknown and eccentric mass properties, explicitly distinguishing support stability from per-agent load feasibility and exposing a failure mode of geometry-centered coordination under hidden eccentricity.
- We propose CoM-HOI, a belief-guided hierarchy that converts short-horizon first-lift responses into a probabilistic CoM belief and uses it to assess transport risk and guide controlled putdown and re-formation.
- We integrate CoM-biased contact allocation, CoM-aligned re-lift initialization, and analytic load-aware rewards. Across hidden-eccentric tasks, CoM-HOI remains competitive at mild eccentricity and delivers increasing gains at larger offsets while maintaining robust transport.

2 Related Work

Cooperative Humanoid-Object Interaction. Cooperative HOI requires coordinated contacts, formation, locomotion, and stabilization. MAPPO (Yu et al. 2022) provides a centralized-training, decentralized-execution foundation, but contact-rich transport introduces physical constraints beyond generic coordination. CooHOI (Gao et al. 2024) transfers carrying skills to multi-agent transport via implicit object-dynamics communication. TeamHOI (Lionar and Lee 2026) supports variable team sizes using teammate tokens, masked AMP, and formation rewards. COLA (Du et al. 2026) studies human-humanoid carrying, while SMPLOlympics (Luo

et al. 2024) provides multi-person sports environments. Existing methods emphasize geometric coordination, but hidden eccentric CoMs can destabilize transport through tipping and uneven loading, leaving interaction-based physical probing and adaptive re-formation underexplored.

CoM Estimation through Physical Interaction. Visual grasping methods increasingly consider mass properties, but often approximate the CoM from object geometry. AnyGrasp (Fang et al. 2023) incorporates a CoM-aware stability score while assuming uniform density, and therefore cannot resolve visually hidden mass eccentricity. Interaction-based methods instead exploit forces, torques, slip, or object motion. (Kanoulas et al. 2018) combine 3D range sensing with wrist force/torque feedback to iteratively adapt grasp poses toward the estimated CoM. (McGovern, Mao, and Xiao 2019) learn CoM prediction through manipulation using reinforcement learning, while (Feng et al. 2020) fuse RGB-D, tactile, and torque signals for slip-guided regrasping. (Hyland, Xiao, and Onal 2023) estimate planar CoM through iterative pushing, and U-GRAPH (Jin, Mo, and Yuan 2025) uses Bayesian uncertainty to select an informative second rotation and fuse multiple force/torque measurements. However, these works mainly address single-arm grasping, regrasping, or planar estimation; they do not connect interaction-derived CoM uncertainty to support stability and per-agent load feasibility.

3 Methodology

Our goal is to enable a team of humanoid agents to cooperatively transport an object whose mass distribution is hidden and potentially eccentric. Unlike geometry-only cooperative carrying, our setting requires agents to infer the latent CoM through interaction and adapt their support configuration accordingly. We formulate the task as a belief-guided hierarchical policy with three stages: interaction-based physical probing (Sec. 3.2), physics-based CoM belief estimation (Sec. 3.3), and belief-guided re-lift and transport (Sec. 3.4).

3.1 Preliminary

We consider N active humanoid agents jointly interacting with a rigid object. The object geometry is observable, and its geometric center is denoted as $g \in \mathbb{R}^2$. The true CoM is denoted as $c^* \in \mathbb{R}^2$, which is hidden from all decentralized actors. Each agent observes its proprioceptive state, object geometry, candidate contact points, teammate cues, and interaction feedback, but does not observe c^* during execution.

Following cooperative humanoid carrying settings, we sample a set of candidate contact points $\mathcal{Q} = \{q_k\}_{k=1}^K$ along the object boundary. Each agent acts independently with a shared policy, while coordination is achieved through object observations, teammate cues, and belief-conditioned inputs. The action of agent i is generated by:

$$a_t^i \sim \pi_\theta \left(a_t^i \mid o_t^i, z_t^{\text{stage}}, z_t^{\text{goal}}, z_t^{\text{belief}}, z_t^{\text{team}} \right), \quad (1)$$

where o_t^i is the local observation, and the additional tokens encode the current task stage, target goal, CoM belief, and team configuration.

As illustrated in Fig. 2, cooperative transport requires satisfying two distinct physical constraints. The first is *support*

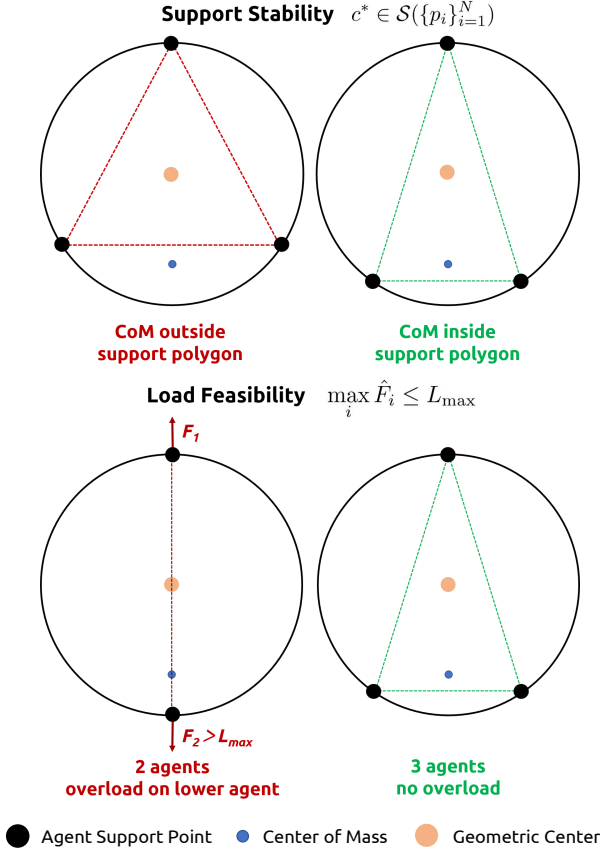


Figure 2: Illustration of two physical constraints in cooperative humanoid transport. Support stability requires the true CoM to lie inside the support polygon induced by agent contacts, while load feasibility requires the predicted load on each agent to remain below its capacity.

stability: the support region formed by active contacts should cover the true CoM to prevent object tipping or contact loss. The second is *load feasibility*: the vertical load assigned to each agent should not exceed its rated load capacity. These two constraints are not equivalent. A formation may provide geometric support for the object, yet still induce severe load imbalance when the CoM is eccentric, overloading agents closer to the heavy side.

Given actual contact locations $\{p_i\}_{i=1}^N$ and predicted vertical loads $\{\hat{F}_i\}_{i=1}^N$, we formalize the two conditions as:

$$c^* \in \mathcal{S}(\{p_i\}_{i=1}^N), \quad \max_i \hat{F}_i \leq L_{\max}, \quad (2)$$

where $\mathcal{S}(p_i = 1^N)$ denotes the support region induced by active contacts, and $L_{\max}$ is each humanoid’s rated load capacity. The first condition captures whether the object is geometrically supported, while the second determines whether the team can transport it without overburdening any agent.

3.2 First Lift as Interaction-Based Physical Probing

Before estimating hidden mass distribution, we reuse a TeamHOI-style cooperative carrying policy as the base con-

troller. This policy provides skills for approaching the object, selecting boundary contacts, establishing hand-object interaction, lifting, and maintaining support. Rather than modifying the action space or policy architecture, we reinterpret initial lifting behavior of the base policy as an interaction-based physical probing process through goal modulation.

Given the visible geometry, the team first adopts a geometry-based support configuration. Let $\mathcal{Q} = \{q_k\}_{k=1}^K$ denote candidate contacts sampled along the object boundary. Since the true CoM is unknown, the initial assignment depends only on the geometric center g and team size:

$$\mathcal{G}^{(0)} = \text{Assign}_{\text{geo}}(\mathcal{Q}, g, N). \quad (3)$$

This assignment serves as the default support pattern before mass-property cues become available.

We then replace the transport goal with a local low-amplitude lift target:

$$x_{\text{tar}}^{(1)} = x_{\text{obj},0}, \quad z_{\text{tar}}^{(1)} = z_{\text{obj},0} + \Delta z_{\text{probe}}, \quad (4)$$

where $x_{\text{obj},0}$ and $z_{\text{obj},0}$ are the initial object position and height, and Δz_{probe} is the probing height. The agents lift the object in place and briefly hold it, preserving the base policy’s cooperative motion while exposing short-horizon responses to hidden eccentricity.

During the probing window $[t_0, t_0 + T_p]$, we record the interaction history:

$$h^{\text{probe}} = \{F_i(t), p_i(t), \phi(t), \omega(t), \delta_i(t), s_i(t)\}_{i=1}^N, \quad (5)$$

where $F_i(t)$ is the measured vertical support force, $p_i(t)$ is the contact location, $\phi(t)$ and $\omega(t)$ denote the object tilt and angular velocity, and $\delta_i(t)$ and $s_i(t)$ describe hand-object displacement and slip. These signals provide short-horizon physical evidence about the hidden CoM.

To make the interaction closer to a quasi-static probing process, we add a lightweight probing regularization term on top of the base policy reward:

$$r_{\text{probe}} = \exp \left(-\lambda_{\phi} \|\phi\|^2 - \lambda_{\omega} \|\omega\|^2 - \lambda_s \sum_i s_i \right). \quad (6)$$

This regularizer suppresses excessive tilt, rotation, and slip during the hold phase, improving CoM belief estimation reliability while retaining compatibility with the base policy.

3.3 Physics-based CoM Belief Estimation

After the interaction-based physical probing stage, we estimate a compact belief over the hidden CoM from short-horizon interaction feedback. The goal of this module is not to reconstruct full object dynamics, but to infer the mass-distribution cue needed for load prediction and team reformation. We maintain a Gaussian belief $b(c) = \mathcal{N}(\mu_c, \Sigma_c)$ initialized around the geometric center g , where μ_c denotes the estimated CoM and Σ_c captures uncertainty.

Under a quasi-static lifting assumption, the CoM is inferred from vertical support forces and contact locations. Using the probing-window averaged support force $\bar{F}_i$ and contact position $\bar{p}_i$ of each agent, we compute:

$$\hat{c}_{\text{ana}} = \frac{\sum_{i=1}^N \bar{F}_i \bar{p}_i}{\sum_{i=1}^N \bar{F}_i}. \quad (7)$$

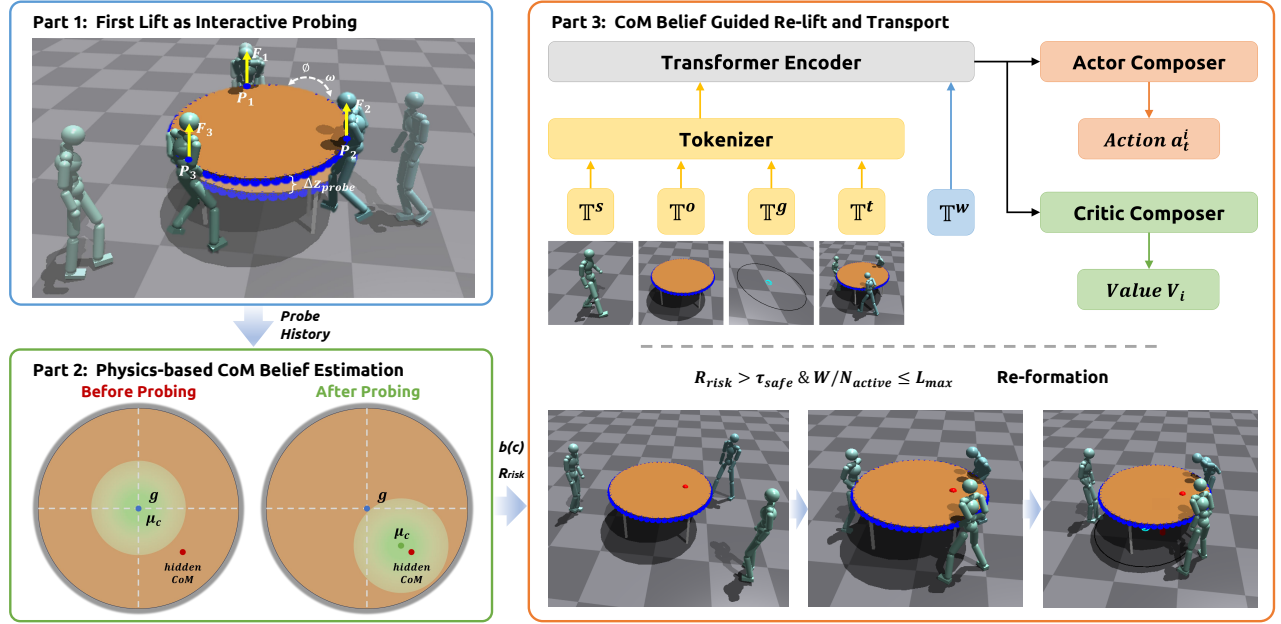


Figure 3: Overview of CoM-HOI. The team performs a low-amplitude lift for interaction-based physical probing, converting short-horizon object responses into a probabilistic CoM belief. Conditioned on this belief, independent actor and critic Transformers predict decentralized actions and state values without parameter sharing. In the three-agent setting, threshold-exceeding risk triggers controlled putdown, belief-guided re-formation, and re-lift; otherwise, the team proceeds with transport.

Here, $\hat{c}_{ana}$ is the estimate from averaged probing measurements, while μ_c is the mean of the final Gaussian belief obtained by aggregating valid estimates over stable hold frames. Because raw forces may be biased by policy-induced overpressing, we use their relative pattern rather than absolute magnitude as the physical cue. We obtain μ_c from stable frames and estimate Σ_c from the temporal dispersion of valid estimates, yielding a belief robust to transient contacts and unstable responses.

Given this belief, we predict the per-agent load distribution under quasi-static force and torque balance. A regularized minimum-norm solver computes vertical loads from the estimated CoM, contacts, and object weight, then projects them onto a nonnegative distribution. The resulting estimate conserves total support force, captures eccentricity-induced load shifts, and is decoupled from over-pressing, serving as the primary signal for overload detection and re-formation.

At the end of probing, the hierarchical manager evaluates whether the current formation is safe using a risk score that combines load feasibility and object stability:

$$R_{risk} = w_{ov} \text{ReLU} \left(\frac{\max_i \hat{F}_i}{L_{max}} - 1 \right) + w_{tilt} \frac{\|\phi\|}{0.3} + w_{slip} \frac{\sum_i s_i}{0.3}. \quad (8)$$

The first term detects predicted overload with respect to the rated load capacity L_{max} , while the remaining terms penalize the object tilt $\|\phi\|$ and aggregate contact slip $\sum_i s_i$. This risk score determines whether the team proceeds with transport or triggers safe putdown and belief-guided re-formation.

3.4 CoM Belief Guided Re-lift and Transport

After probing, the hierarchical manager uses R_{risk} as a stage-transition signal. If $R_{risk} \leq \tau_{safe}$, the team retains its formation and proceeds with transport; otherwise, it safely puts down the object before belief-guided re-formation and re-lift. We consider fixed three-agent teams whose nominal load capacity is sufficient for all evaluated objects, i.e., $W/3 < L_{max}$. Accordingly, it adapts the support configuration without changing the team size. Recruitment for capacity-infeasible settings is outside the scope of this study.

Belief-conditioned policy. As shown in Fig. 3, we retain the TeamHOI tokenizer–Transformer (Vaswani et al. 2017) backbone. For each agent, self-proprioception, object information, task goals, and teammate cues are encoded as T^s , T^o , T^g , and T^t . A learnable pooling token T^w aggregates T^s , T^o , and T^g through self-attention, while cross-attention incorporates the variable-size teammate-token set T^t .

Before constructing T^o , we append an eight-dimensional CoM-belief block to the object observation. It encodes the estimated CoM direction and magnitude, uncertainty, agent-specific load ratio $\hat{F}_i/L_{max}$, active and maximum team sizes, and a capacity-feasibility indicator. In the fixed three-agent setting, the indicator records nominal capacity status without changing team size. The actor uses only the estimated belief to produce a_t^i , whereas the critic outputs V_i and may access the true CoM c^* during centralized training. We warm-start shape-compatible parameters from the base policy and reinitialize only the modified object-tokenizer input layer, preserving learned locomotion and carrying skills for belief-conditioned re-formation.

Belief-guided initialization. Re-formation biases candidate contacts toward the estimated heavy side. The CoM bearing is $\psi_c = \text{atan2}(\mu_{c,y} - g_y, \mu_{c,x} - g_x)$. For a candidate with boundary angle θ_k , let α_k denote its wrapped displacement from ψ_c . Its transformed angle θ'_k reduces this displacement by $\kappa_g \sin \alpha_k$, where κ_g controls grasp-density bias. This transformation concentrates contacts near the heavy side while preserving boundary coverage. Under high belief uncertainty, the bias is attenuated to prevent unreliable estimates from producing overly concentrated allocations.

We further initialize the re-lift formation so that its support centroid shifts toward μ_c . For agents placed on a rim of radius R_{rim} with angular positions $\{\beta_k\}_{k=1}^N$, the formation is selected to approximately satisfy:

$$\frac{1}{N} \sum_{k=1}^N \cos(\beta_k - \psi_c) = \frac{\|\mu_c - g\|}{R_{\text{rim}}}. \quad (9)$$

This provides the low-level policy with a physically meaningful asymmetric initialization, avoiding reward optimization to discover the shift from the geometry-centered formation. During re-formation and re-lift, uniform angular-spacing and geometry-centered coverage targets are replaced by stage-gated, belief-conditioned targets that preserve the probing lift. The geometry bias narrows the controller’s search space and facilitates policy optimization.

Load-aware re-formation objective. The re-formation policy is trained with PPO (Schulman et al. 2017) and AMP while retaining the original TeamHOI task and motion-style rewards. We augment the base objective with support-margin, overload, stability, over-press:

$$r_i = r_{\text{TeamHOI},i} + \lambda_m r_{\text{margin}} + \lambda_c r_{\text{cap}} + \lambda_s r_{\text{stable}} + \lambda_o r_{\text{over},i}. \quad (10)$$

Here, r_{margin} encourages the support region to contain the estimated CoM belief, while r_{cap} penalizes predicted loads above L_{max} . The stability term r_{stable} suppresses object tilt, angular motion, and contact slip, whereas $r_{\text{over},i}$ uses measured contact forces to discourage controller-induced over-pressing rather than replace the analytic load prediction. Together, the initialization and reward design couple support geometry with local agent-wise load feedback. After a stable re-lift, the belief-conditioned policy transports the object to the target and performs the final putdown.

4 Experiments

We organize the experiments around three questions:

- **Q1: Problem necessity** examines how hidden CoM offsets affect conventional cooperative transport.
- **Q2: Belief and adaptation** evaluates the accuracy of first-lift CoM estimation and the effectiveness of belief-guided re-formation across varying eccentricities.
- **Q3: Component contributions** isolates the roles of key components through controlled ablations across eccentricity levels.

All reported comparisons use frozen-policy evaluation under matched protocols so that the observed differences reflect test-time behavior.

4.1 Experimental Setup

Humanoid and object models. We adopt the MuJoCo (Todorov, Erez, and Tassa 2012) humanoid model with simplified ball hands and use the same three table geometries as TeamHOI: square (1.60 m $\times$ 1.60 m), rectangular (2.00 m $\times$ 1.20 m), and round (diameter 2.00 m). The table-top height is fixed at 0.82 m, slightly below the default hand height, so the experiments primarily evaluate cooperative coordination rather than complex grasping. The table mass ranges from 60 to 75 kg depending on the shape, and each humanoid has a rated load capacity of $L_{\text{max}} = 300$ N. All reported experiments fix the active team size to $N = 3$. This setting isolates support re-formation under hidden mass properties from team-size adaptation.

Hidden-Eccentric Cooperative Transport. We extend the TeamHOI table-carrying task with a latent horizontal CoM offset. The visible geometry remains unchanged, preventing inference of the offset from the object mesh. We use round, square, and rectangular tables and define normalized eccentricity as $\eta_{\text{CoM}} = \|c^* - g\| / r_{\text{obj}}$. The evaluation uses three discrete offset magnitudes rather than interval or disk sampling: Easy ($\eta_{\text{CoM}} = 0.30$), Medium ($\eta_{\text{CoM}} = 0.45$), and Hard ($\eta_{\text{CoM}} = 0.60$). For each episode, the offset direction is sampled uniformly from $[0, 2\pi)$, with magnitude fixed by the selected level. For non-circular tables, r_{obj} is the center-to-boundary distance along the sampled direction, ensuring comparable fractional eccentricity across shapes. All three shapes are sampled equally at each difficulty level.

Implementation details. Experiments are conducted in Isaac Gym (Makoviychuk et al. 2021) on a single NVIDIA H100 GPU. The actor and critic use independent TeamHOI tokenizer–Transformer architectures without parameter sharing. Each network uses 64-dimensional tokens and six alternating self- and cross-attention layers. The eight-dimensional belief block is appended to the object observation; the actor receives only the estimated CoM belief, whereas privileged true CoM information, when enabled, is restricted to the critic or an explicitly labeled Oracle actor. Training combines PPO with AMP following the corrected two-stage TeamHOI curriculum. Because the belief input changes the object-tokenizer dimension, re-formation training is warm-started from the pretrained carrying policy by loading shape-compatible parameters and reinitializing only the expanded input layers. Training uses 2048 parallel environments.

4.2 Evaluation

Baselines: We compare against two representative cooperative humanoid-object interaction baselines. TeamHOI uses explicit teammate cues and a geometry-centered formation through a decentralized Transformer policy, but does not estimate hidden mass properties or adapt contacts to the CoM. CooHOI*-3 is a three-agent adaptation of CooHOI under the TeamHOI evaluation protocol. It coordinates through shared object dynamics, removes teammate tokens, and assigns contacts using a predefined geometric mapping. Both baselines are retrained on the same quality objects with hidden-CoM distribution and evaluated with identical object assets, episode budgets, and metric implementations.

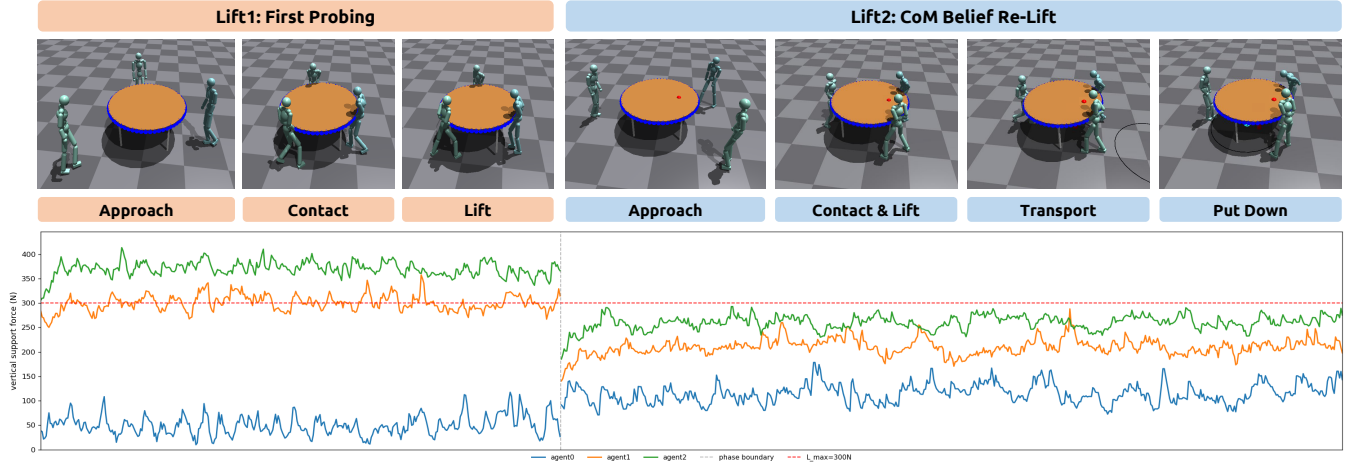


Figure 4: Load redistribution through CoM-aware re-formation. The team performs a probing lift from a geometry-based formation before adapting its support configuration using the estimated CoM belief. Force trajectories show that CoM-guided re-formation redistributes agent loads, enabling more balanced cooperative transport under hidden eccentric mass properties.

Method	Agents Formation	Capabilities			SR (%) $\uparrow$ / d (m) $\downarrow$			t_{coop} (%) $\uparrow$			$ J $ (m/s ³) $\downarrow$		
		Dyn.	Mate	CoM	E30	M45	H60	E30	M45	H60	E30	M45	H60
CooHOI*-3	Predefined	✓	–	–	59.6/6.70	34.9/6.74	17.5/6.77	92.7	84.6	67.1	64.1	69.2	85.8
TeamHOI	Geometric	✓	✓	–	97.5/0.23	84.6/0.81	59.2/2.36	98.2	89.2	69.1	51.8	56.4	70.5
CoM-HOI(Ours)	Belief-guided	✓	✓	✓	96.9/0.33	94.3/0.52	81.8/1.32	98.9	97.5	85.6	50.4	51.2	63.9

Table 1: Comparison with cooperative humanoid-object transport baselines. Main comparison at fixed $N = 3$. E30, M45, and H60 denote $\eta_{\text{CoM}} = 0.30, 0.45, 0.60$. Dyn., Mate, and CoM denote object-dynamics coordination, explicit teammate cues, and CoM-belief-guided adaptation. All entries are computed using the same frozen-policy evaluation protocol.

Metrics: The main quantitative comparison reports four complementary metrics. Higher SR and t_{coop} are preferred, whereas lower d and $|J|$ indicate better performance.

- **Success rate (SR)** is the percentage of episodes in which the object is transported to the target and stably placed within a distance tolerance of $\epsilon_d = 0.3$ m.
- **Target distance (d)** is the Euclidean distance between the object center and the target at stable termination or episode timeout, directly measuring transport accuracy.
- **Cooperative time (t_{coop})** is the proportion of transport frames during which all active humanoids maintain contact with at least one candidate contact point, reflecting sustained team participation.
- **Mean absolute jerk ($|J|$)** is computed from the third temporal derivative of the candidate contact-point trajectories during transport and evaluates motion smoothness.

Results: For *Q1: Problem necessity*, Fig. 1 evaluates the frozen TeamHOI base policy across hidden CoM offsets and team sizes. The geometry-centered policy performs reliably for symmetric objects, but its transport success decreases and its motion becomes less smooth as the hidden CoM moves away from the geometric center. Increasing the team size partially mitigates this degradation by providing addi-

tional support contacts, yet does not recover the centered-object performance under strong eccentricity. Team-size redundancy therefore alleviates, but does not resolve, the mismatch between visible geometry and latent mass distribution. These results expose a limitation of geometry-centered coordination and motivate interaction-based physical probing followed by CoM-aware support adaptation.

For *Q2: Belief and adaptation*, Fig. 5 evaluates whether uncertainty ranks CoM-belief reliability. Across eccentricities, low- σ estimates are better aligned with the ground-truth direction than high- σ estimates. Directional accuracy also improves as CoM offset increases, indicating that stronger mass asymmetry generates more informative interaction responses during the probing lift. This trend reveals a distinction between observability and control difficulty: larger offsets make the hidden CoM direction easier to identify while making stable transport mechanically harder. These results support using σ_c to prevent unreliable directional corrections from propagating into downstream formation decisions.

Fig. 4 further connects belief estimation to physical adaptation. Per-agent loads are measured directly in simulation by summing the vertical contact-force magnitudes over both hand bodies, suppressing solver noise and impulse spikes through thresholding and clipping, and applying an exponen-

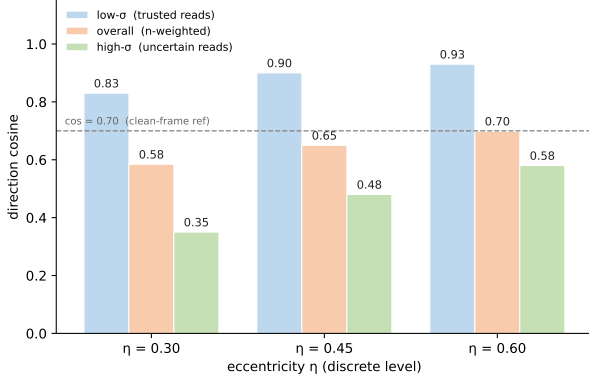


Figure 5: Uncertainty-gated CoM-direction estimation. Direction cosine between estimated and ground-truth CoM directions across three eccentricity levels. The σ -gate separates low- σ trusted estimates from high- σ uncertain estimates, with sample-weighted overall results for comparison. The dashed line denotes the clean-frame reference.

tial moving average to recover the quasi-static level. During the geometry-based probing lift, the eccentric load concentrates on agents nearest the hidden CoM, while another agent remains lightly loaded. Belief-guided re-formation shifts the support polygon toward the estimated heavy side, reducing dominant loads while increasing the contribution of the previously underused agent. Because the object weight is unchanged, this pattern reflects redistribution of support effort rather than a uniform reduction in measured force. The estimated belief therefore provides sufficient directional information to produce a more load-feasible support arrangement.

The task-level results in Table 1 are consistent with this physical interpretation. Under mild eccentricity, CoM-HOI remains comparable to TeamHOI, indicating that belief-conditioned coordination does not materially disrupt an already suitable geometric formation. As eccentricity increases, the geometry-centered baselines deteriorate more rapidly, whereas CoM-HOI better preserves transport success, cooperative contact, target accuracy, and motion smoothness. At H60, CoM-HOI achieves an 81.8% success rate, surpassing the strongest baseline TeamHOI (59.2%) by 22.6 percentage points. Its concurrent improvements in the remaining metrics indicate that the success-rate gain is accompanied by sustained team participation, more accurate placement, and smoother transport. Overall, CoM-HOI reduces the sensitivity of cooperative transport to hidden mass eccentricity by coupling uncertainty-aware CoM estimation with adaptive support re-formation.

4.3 Ablation Study

To answer *Q3: Component contributions*, we remove the σ -Gate and risk-triggered re-formation individually while retaining the same probing procedure, CoM estimator, policy backbone, and frozen-policy evaluation protocol.

Table 2 shows that both components become increasingly important as CoM shift grows. Under mild eccentricity, the geometry-centered formation remains largely feasible, so un-

Method	SR (%) $\uparrow$ / d (m) $\downarrow$	t_{coop} (%) $\uparrow$	$ J $ (m/s ³) $\downarrow$
w/o σ -Gate	95.6 / 0.41	97.8	62.1
w/o Reconfig	96.2 / 0.36	98.2	54.3
Ours	96.9 / 0.33	98.9	50.4
w/o σ -Gate	73.2 / 2.31	79.6	71.5
w/o Reconfig	82.5 / 1.28	88.4	57.8
Ours	94.3 / 0.52	97.5	51.2
w/o σ -Gate	42.1 / 3.75	64.9	89.5
w/o Reconfig	63.5 / 1.95	78.5	72.8
Ours	81.8 / 1.32	85.6	63.9

Table 2: Ablation experiments. *w/o σ -Gate* removes confidence-based filtering and uses the estimated CoM belief regardless of uncertainty. *w/o Reconfig* retains physical probing and uncertainty gating but disables risk-triggered safe shutdown and re-formation. The row blocks, from top to bottom, correspond to $\eta_{\text{CoM}} = 0.30, 0.45$, and 0.60 . All variants follow the same frozen-policy evaluation protocol.

certainty gating and re-formation primarily refine interaction quality rather than determine task completion. As the mass distribution becomes more asymmetric, however, inaccurate directional estimates and unsuitable initial contacts have greater physical consequences. Without the σ -Gate, uncertain beliefs can propagate into misguided formation updates, causing the team to adapt in an incorrect direction. Without re-formation, the team cannot revise an initially unfavorable support configuration even when estimated risk indicates uneven loading or instability. The stronger degradation caused by unfiltered beliefs further suggests that unreliable adaptation can be more harmful than conservatively retaining the current formation. These results highlight the complementary roles of the two components: uncertainty gating controls whether the belief should influence downstream decisions, while re-formation provides the physical correction needed when the current support arrangement is unsafe. Combining both mechanisms enables CoM-HOI to maintain reliable and coordinated transport as mass eccentricity increases.

5 Conclusion

We presented CoM-HOI, enabling belief-guided cooperative humanoid-object transport under uncertain mass properties. CoM-HOI treats a low-amplitude lift as interactive physical probing, converting short-horizon responses into a probabilistic CoM belief. Using this belief, the manager evaluates support stability and load feasibility, retaining feasible formations or triggering controlled shutdown, CoM-aware re-formation, and re-lift. Integrating belief estimation, formation adaptation, and decentralized whole-body control complements geometric coordination with latent mass reasoning. Across hidden-eccentric tasks, it remains competitive under mild eccentricity and maintains robust transport with greater advantages as the CoM offset grows. Estimation and ablation results further support the roles of interaction-based physical probing and belief-guided adaptation.

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